A11yPreserve: An Uncertainty-Aware Conformance Suite for DOCX-to-PDF Accessibility Preservation

Anonymous for review

***Abstract*—Accessible source documents can contain author-supplied information such as heading structure, alternative text, language metadata, list semantics, table relationships, and footnote associations. When a document is converted, inspecting only the destination does not establish whether those source properties survived. We present A11yPreserve, an uncertainty-aware conformance suite that freezes controlled source contracts, independently checks their DOCX representations, converts identical sources through real pipelines, extracts destination PDF structure through multiple evidence paths, and compares source and destination properties with provenance-traceable evidence. The atomic benchmark contains 26 controlled DOCX fixtures, two per selected feature family, and 52 fixture–pipeline cases from LibreOffice and the Google Docs DOCX-to-PDF conversion pipeline. Its evidence-aware results are 18 verified preserved, 4 observed partial, 7 altered, 4 confirmed lost, 18 unresolved equivalence, and 1 measurement error. The four confirmed-loss rows are the Core and Variant B instances of inline-language and footnote association under the tested Google Docs pipeline. A separate 39-run Core repeatability study found stable classifications for all 13 fixtures, with byte-identical PDFs across the three recorded runs. A separate three-document integrated sanity check shows that the protocol remains usable when existing feature contracts coexist. Results are feature-specific and are not used to rank converters globally. The study measures machine-verifiable source-information preservation, not complete accessibility or disabled-user impact.**

***Index Terms*—document conversion, accessibility preservation, DOCX, PDF, tagged PDF, differential testing, document structure**

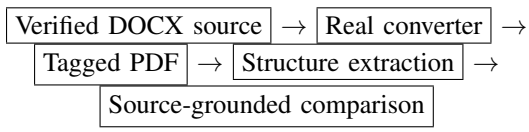
I. Introduction

Authors can encode accessibility information directly in source documents: a heading is more than large text, an image may carry an author-written alternative, a run may identify a language change, and a table may identify header relationships. These properties matter because the source document is often exported, uploaded, or converted before it reaches readers. The resulting file may contain some accessibility structure, but that fact alone does not show whether the information intentionally supplied by the author survived the transformation.

This distinction is recognized in accessibility guidance. ATAG 2.0 addresses preservation during content transformations when equivalent output mechanisms exist [1]. Revised Section 508 includes a requirement to preserve information provided for accessibility during format conversion to the extent supported by the destination format [2]. EN 301 549 likewise specifies preservation of accessibility information in transformations [3]. These requirements motivate a practical question: how can preservation be tested against known source ground truth rather than inferred from the destination alone?

Prior work has studied office-document conversion, PDF-to-HTML accessibility, automated PDF remediation, and destination accessibility evaluation. Roig and Ribera examined office-to-EPUB conversion [4]; SciA11y and related Paper-to-HTML work reconstruct accessible representations from scientific PDFs [5]; iTagPDF uses source cues to improve PDF tagging [6]; and recent benchmarks evaluate PDF accessibility or conversion services [7]–[10]. A11yPreserve complements these efforts by beginning with controlled DOCX documents whose target accessibility properties are explicitly verified, then measuring whether those properties survive two real DOCX-to-PDF pipelines. The contribution is the controlled source-to-destination preservation protocol and empirical benchmark, not a claim that accessible conversion has not been studied.

We present A11yPreserve as a source-contract conformance protocol, benchmark, and empirical study. The workflow is:



In testing terms, the source contracts are feature-specific test oracles, the fixtures are controlled test inputs, and each conversion pipeline is a system under test. Destination extraction supplies observations; the comparison rules make the oracle decision and retain evidence paths. When the available evidence cannot establish an accepted cross-format representation, the result is explicitly unresolved rather than forced into pass or fail.

The atomic benchmark contains 26 controlled DOCX fixtures, two per selected feature family, and 52 fixture–pipeline cases: 26 from LibreOffice and 26 from the Google Docs DOCX-to-PDF conversion pipeline. All atomic outputs were parseable, non-empty, tagged, and passed the predefined structural output checks. The evidence-aware results are 18 VERIFIED PRESERVED, 4 OBSERVED PARTIAL, 7 ALTERED, 4 CONFIRMED LOST, 18 UNRESOLVED EQUIVALENCE, and 1 MEASUREMENT ERROR. A separate Core repeatability study contains 39 Google conversions (13 fixtures across three runs); all 13 tested classifications were stable and the PDFs were byte-identical within the recorded environment. A separate secondary sanity check evaluates three project-generated multi-feature integrated documents using the same contracts and pipelines; it is not merged into either denominator. The four confirmed-loss rows are the Core and Variant B instances of inline language and footnote association in the tested Google Docs pipeline. These findings are not a product

ranking and do not estimate the prevalence of conversion problems in arbitrary documents. Earlier Core runs remain archived for provenance rather than being silently overwritten.

The primary research question is: *Does accessibility information present in a source document survive document conversion?* The paper makes three methodological contributions and one bounded empirical application:

1) versioned, feature-specific source contracts for controlled DOCX accessibility properties;
2) a separately implemented source oracle and destination-evidence protocol that separates verified equivalence, observed difference, unresolved equivalence, confirmed absence, and measurement failure;
3) a reusable conformance-suite design for supplying new pipeline outputs, with calibrated differential comparison, evidence certificates, and provenance; and
4) an empirical application to a 52-case controlled atomic benchmark, a separate 39-run Google repeatability study, and a three-document integrated sanity check, reporting preservation behavior only for the tested fixtures, documents, and pipeline conditions.

The paper deliberately makes no universal claim about accessibility loss during conversion, does not rank either converter globally, and does not treat the results as real-world prevalence. It asks a simpler question: when accessibility information is known to exist before conversion, did that information survive?

II. Background and Related Work

A. Preservation as an accessibility requirement

Accessibility information is often authored in one representation and transformed into another. ATAG 2.0 explicitly addresses restructuring and recoding transformations: when equivalent mechanisms exist in the output technology, accessibility information should be preserved or the author should be warned or prompted to check it [1]. Its text-alternative criterion is especially close to the F02 contract. The U.S. Revised Section 508 standards similarly state that authoring tools should preserve information required for accessibility during format conversion to the extent supported by the destination format [2]. EN 301 549 contains a corresponding preservation-of-accessibility-information requirement for transformations [3]. These requirements motivate testing preservation, but they do not predict the outcomes of the converters evaluated here.

B. Accessibility-aware document conversion

Roig and Ribera evaluated office-document conversion to EPUB and reported accessibility-preservation problems in that setting [4]. Their work is an important precedent for studying accessibility effects of conversion. A11yPreserve differs in experimental scope and instrumentation: it uses frozen native DOCX source contracts, real DOCX-to-PDF pipelines, destination structure extraction, and provenance-traceable fixture-level differential records. The distinction is one of scope and method, not a claim that prior work ignored preservation.

C. Testing foundations for transformation oracles

The testing contribution is grounded in the oracle problem: a test can execute and produce an output even when deciding whether that output is correct is difficult [11]. A11yPreserve uses a specification-derived, feature-level oracle: the DOCX contract states the intended accessibility property, while the comparator must decide whether a PDF representation is equivalent rather than merely textually similar. The conversion is therefore a cross-format oracle problem, not a claim to introduce oracle theory. The paired source/output analysis is also related to differential testing, but the central relation is contract-based rather than a majority vote among converters. Metamorphic testing is a neighboring strategy for oracle construction [12]; it is not the primary oracle here because the source contracts provide explicit expected properties and the DOCX-to-PDF transformation changes representation. Format-conversion scholarship likewise cautions that successful conversion requires a declared notion of what must be preserved, not only syntactic validity [13]. These foundations position A11yPreserve as a domain-specific empirical testing protocol that makes representability and abstention explicit.

D. Accessible representation generation and remediation

SciA11y converts scientific papers from PDF to accessible HTML and combines extraction with navigation-oriented rendering for blind and low-vision readers [5]. This work primarily creates a more accessible destination representation. iTagPDF uses source-document cues and visual PDF analysis to automate tagging, reading order, and content-specific metadata such as alt text and formula structure [6]. Its source-aware remediation objective is closely related to A11yPreserve's use of source information, but the experimental question differs: iTagPDF generates or improves PDF metadata, whereas A11yPreserve measures whether author-provided source information survived an existing conversion pipeline.

E. PDF conversion and accessibility evaluation benchmarks

Recent work also benchmarks destination conversion or evaluation. Kumar, Padath, and Wang introduce a benchmark for assessing automated and LLM-based PDF accessibility evaluation across criteria including alternative text, reading order, tagging, tables, and hyperlinks [7]. DAISY's 2026 benchmark compares seven services that convert PDFs into accessible documents against a shared quality framework [8]. DocAccessible's public research program reports source-grounded PDF-to-HTML conversion and fidelity benchmarking with frozen evidence [9]. Its current Google Docs guidance also emphasizes that tagged export requires post-export checks of language, title, table headers, and reading order, and does not itself assert WCAG or PDF/UA conformance [10]. These are important adjacent efforts. A11yPreserve's narrower contribution is to start from controlled DOCX accessibility ground truth and compare the same source semantics across real DOCX-to-PDF pipelines, while separating preservation fidelity from destination-only accessibility evaluation.

Vendor documentation provides useful format context as well. Microsoft's current Word documentation maps source constructs such as language spans, footnotes, equations, captions, hyperlinks, and decorative content to PDF structure, including `/Lang`, `/Note`, `/Formula`, `/Caption`, and artifacts [14]. We use such documentation to define representability and scope, not as comparative evidence about the tested pipelines.

The resulting contribution boundary is therefore modest: not the invention of accessibility-conversion benchmarking, but a source-grounded DOCX-to-PDF preservation conformance design with explicit contracts, independently checked source hashes, calibrated feature-level evidence statuses, and a reproducible evidence chain. The contribution is the controlled source-to-destination preservation protocol and empirical benchmark; it is not a claim that prior work lacked accessible-conversion studies.

III. A11yPreserve Method

A11yPreserve treats accessibility preservation as a source-to-destination comparison problem. The method begins with a document whose target accessibility properties are known, runs that exact source through a named conversion pipeline, extracts accessibility-relevant destination structure, and classifies the correspondence. This framing is intentionally narrower than asking whether a PDF is accessible in every respect. A destination-only checker can report tags or metadata that are useful in their own right, but it cannot by itself establish whether an author-provided source property was retained.

In software-testing terms, each source contract is a specification-derived oracle for one intended semantic property. The oracle decision is cross-format: an expected property in OOXML is compared with an accepted representation in PDF after the conversion system under test has changed the document representation. This is an application of oracle-aware and differential testing ideas, not a claim to invent them. The decision procedure records the accepted equivalent representations explicitly; when the destination evidence cannot establish equivalence without assuming a particular encoding, it abstains as UNRESOLVED EQUIVALENCE rather than treating an extractor omission as a failure or success. The resulting outcome is a test result for a fixture–pipeline case, not a complete accessibility judgment.

A. Source fixtures and ground truth

The frozen Core corpus contains 13 small DOCX fixtures. The fixtures cover heading hierarchy, image alternative text, list semantics, table/header structure, document language, inline language changes, decorative image state, hyperlinks, document title and metadata, multi-level table headers, footnote association, equation/math semantics, and figure/caption association. They were created by controlled generation scripts using `python-docx` together with explicit WordprocessingML constructs where required. Each Core fixture isolates a target property and avoids unnecessary layout complexity. Variant Set B adds one pre-specified second controlled instance for each of the 13 feature families, yielding two controlled instances per selected family in the atomic benchmark. F13 was reserved for reading order but deferred because reliable automated source-to-PDF equivalence could not be established; the frozen identifiers therefore remain F01–F12 and F14, without renumbering.

Each fixture has a machine-readable preservation contract under `contracts/`. The frozen source-grounding chain used three artifacts: the source manifest, raw OOXML inspection, and the source extractor's accessibility intermediate representation (ASIR). The strengthening pass added a fourth, separate audit path: a separately implemented source oracle performs fresh ZIP/XML assertions without importing ASIR, fixture-generation, manifest-generation, or comparator code; its per-fixture certificates are retained under `evidence/source/`. OOXML checks inspect the relevant document parts, including document XML, run properties, numbering, relationships, and the footnotes part where applicable. The ASIR remains an implementation-grounded oracle, not an independent implementation or independent human expert validation: it is part of the same analysis toolchain. Agreement among the frozen manifest, raw OOXML, separately implemented oracle, and ASIR provides a stronger cross-check against extractor-only errors, but does not remove the limitation that no external expert panel validated every contract. This establishes encoded source ground truth; it does not claim validation by disabled users. The corpus manifest records file size, SHA-256, feature family, and verification status. The separately implemented OOXML oracle agreed with all 13 frozen source contracts.

B. Test-design and adequacy boundary

The suite makes a deliberately qualitative adequacy claim at four levels. It covers 13 selected accessibility feature families; two controlled source instances per family in the atomic arm; two named conversion conditions; and six evidence-aware oracle outcomes, including abstention and measurement failure. The contracts are also designed to expose four observable classes of transformation behavior: author-value mutation, component deletion, structural or association substitution, and insufficient evidence for a decision. This coverage is a design boundary for the benchmark, not a standards-completeness claim, mutation score, or exhaustive fault model. It makes explicit what a new converter adapter can be tested against while leaving broader adequacy as future work.

C. Conversion pipelines

The first condition used LibreOffice 26.2.6.3, build `8221e31b3ac356a1623c672912a3d2b492f7e3d1`, on Windows `10.0.26200.0`. Conversion used native headless `soffice.com writer_pdf_Export` with `UseTaggedPDF=true`, `PDFUACompliance=true`, and `SelectPdfVersion=1`. The second condition used an authenticated Google Docs web workflow on the same Windows environment with Chrome 153.0.8010.53: upload the exact frozen DOCX and use the Google Docs PDF export workflow. For the frozen Core and its two repeatability

runs, the visible workflow was *File → Download → PDF Document (.pdf)*. For Variant Set B, the authenticated upload succeeded for all 13 files, but the visible download action did not register a Chrome download after retries; the saved PDFs were obtained through the in-session Google Docs export backend. This route deviation is recorded in the conversion report and was not a print-to-PDF, screenshot, manual content repair, or post-export accessibility repair. The observed Google PDF producer was `Skia/PDF m156`; the cloud backend version cannot be fully pinned. The official LibreOffice documentation describes its PDF/UA export option [15]; Google's download documentation describes the selected-file-type download workflow [16].

The Google condition is described throughout as the *Google Docs DOCX-to-PDF conversion pipeline*. It includes DOCX import, Google's internal document representation, and PDF generation. The backend version cannot be fully pinned, so the experiment date matters and a rerun may change after a service update; the workflow, browser, operating system, observed producer, and environment were recorded. The experiment does not isolate which internal stage caused an observed change. Microsoft Word was not included in the denominator because it was not a completed primary pipeline.

D. Destination extraction and validation

All 52 atomic outputs were checked for existence, non-zero size, parseability, expected page count, structural tagging, and a PDF structure tree. The extractor inspected PDF structure elements and roles relevant to the fixture contracts: headings, figures and `/Alt`, list containers and `/ListNumbering`, tables and header cells, `/Lang`, link annotations, metadata, `/Note`, `/Formula`, and `/Caption`. Evidence was retained as structured JSON and raw object observations. Poppler `pdftotext` was used as an independent secondary check when marked-content text was not recoverable from the structural parser.

The extractor is deliberately not treated as infallible. For example, the LibreOffice F11 output contains a `/Note` structure, but the available extraction evidence does not establish the note-body association. That case is retained as MEASUREMENT ERROR rather than converted into a positive or negative preservation claim. For F06, F11, and the difficult partial cases, an independent triangulation path uses PyMuPDF page/text/link extraction and serialized-object inspection in addition to pypdf structure traversal. This reduces, but does not eliminate, parser uncertainty.

E. Differential comparison

The `a11ydiff` implementation loads the source ASIR and destination extraction for each fixture/pipeline pair. It records the source value, destination value, destination representation, evidence paths, source/output hashes, confidence, and a classification. The feature contracts and classification rules were fixed before the final conversion results were reviewed. A separate synthetic calibration set exercises positive, negative, and alternate-equivalent destination controls. Across 27 diagnostic controls, 12 were correct determinate results, 6 were correctly retained unresolved, 0 produced false positives, 2 were false negatives, and 7 were other unresolved or indeterminate cases. These counts are not sensitivity, specificity, or population estimates; the alternate-encoding misses motivate retaining unresolved equivalence as a first-class outcome. The implementation also records a secondary destination-accessibility axis. The final evidence-aware result labels are defined below.

Verified preserved: The destination expresses the contracted property exactly or through a verified equivalent representation, with independent source and destination evidence.

Observed partial: Some contract components survive and at least one required representable component is directly demonstrated to differ or be absent.

Altered: The feature remains present, but an author-relevant value changes, such as alt-text wording, locale specificity, or title metadata.

Confirmed lost: The source property is valid and representable in the destination, but the required destination representation is absent and independent checks support that absence.

Unresolved: Relevant destination structure exists, but available evidence cannot establish whether the source contract survived through an alternate PDF representation; absence has not been independently demonstrated.

Measurement error: The output is valid and relevant structure is present, but the available evidence cannot resolve the contract reliably.

An earlier result set used five categories: PRESERVED, PARTIALLY_PRESERVED, ALTERED, LOST, and MEASUREMENT_ERROR. The evidence-aware taxonomy was introduced after methodology review rather than preregistered; unchanged stored evidence was systematically reclassified under the final decision procedure, and label-hidden reclassification reproduced all 26 frozen Core labels. This is a decision-procedure reproducibility check, not independent external adjudication. The final rules were frozen before Variant Set B outputs were collected. The source and destination files did not change.

F. Independent verification and freezing

Non-preserved cases were reviewed against the manifest, raw OOXML, independent source certificates, source ASIR, output validation, primary PDF extraction, full structure traversal, and independent parser/object checks where appropriate. The Core and Variant B Google F06/F11 rows were retained as confirmed loss only after two destination evidence paths corroborated absence of the required machine-verifiable representation; visible text survival is not treated as preservation of the associated structure. Source fixtures and all new PDF outputs were SHA-256 frozen. The earlier evidence-aware result file remains unchanged; the paper reports the 52-case atomic extension, the separate 39-run repeatability study, and the separate integrated sanity check without merging their denominators.

IV. Experimental Design

A. Research questions

The experiment addresses three questions:

1) **RQ1:** To what extent do the tested DOCX-to-PDF conversion pipelines preserve accessibility information present in controlled source documents?
2) **RQ2:** Which tested accessibility properties are verified as preserved, observed as changed or lost, or unresolved under the available cross-format evidence?
3) **RQ3:** How do preservation outcomes differ between the tested LibreOffice and Google Docs conversion pipelines?

Validator comparison is not a fourth core question. The project did not obtain a systematic PAC, Acrobat Accessibility Checker, or veraPDF dataset for all 52 atomic outputs, so destination-only validator comparison is future work rather than a central result.

B. Benchmark composition

Table I summarizes the 13 Core feature contracts. Variant Set B provides one additional pre-specified controlled instance for each contract, so the atomic benchmark contains 26 fixtures and $26 \times 2 = 52$ fixture–pipeline cases. The fixtures are intentionally controlled rather than randomly sampled. F13 was reserved for reading order but deferred because reliable automated source-to-PDF equivalence could not be established; the frozen corpus therefore uses F01–F12 and F14 without renumbering. The table's contract descriptions state what is compared; they do not imply that the feature families have equal complexity or equal user impact.

C. Converter conditions and provenance

The LibreOffice workflow was deterministic and local. The Google Docs workflow was authenticated and browser-mediated, so its service backend was not independently version-pinned. The experiment date, browser version, observed PDF producer, source hashes, output hashes, file sizes, and validation records were retained in the automation log. All outputs were one-page PDFs in this corpus and passed the output-validation checks.

The experiment's primary unit is one fixture/converter pair. The 26 fixtures are controlled atomic cases, not a representative sample; features have unequal complexity and the two instances per selected family do not form a population sample. Selection was purposive, guided by accessibility requirements, common machine-representable DOCX/PDF structures, conversion relevance, and feature-specific source-to-destination mappings. The suite is not an exhaustive adequacy argument: no formal coverage metric or complete fault model is claimed. Counts and feature-level matrices are therefore primary, while percentages are not used as headline results. No inferential statistics, significance tests, prevalence estimates, or weighted overall converter score are justified. The 39-run Google repeatability arm and the separate secondary sanity check use distinct denominators; the latter contains three project-generated multi-feature integrated documents and its property-level observations are not merged into the atomic benchmark.

D. Outcome accounting

The atomic evidence-aware set contains 18 VERIFIED PRESERVED, 4 OBSERVED PARTIAL, 7 ALTERED, 4 CONFIRMED LOST, 18 UNRESOLVED EQUIVALENCE, and 1 MEASUREMENT ERROR case. Thus 52 atomic cases were attempted and successfully converted, with one case remaining a measurement error. The earlier Core baseline remains separately recorded as 11 verified preserved, 3 observed partial, 3 altered, 2 confirmed lost, 6 unresolved, and 1 measurement error. No percentages are used because the controlled feature units have unequal complexity and do not support prevalence interpretation.

E. Repeatability arm

The repeatability study reused the 13 frozen Core DOCX sources and produced two new Google conversions per source, for 39 total Google runs including historical Run 1. Run 2 and Run 3 were separate uploads under the recorded workflow. The three PDFs were byte-identical for all 13 fixtures, and the corresponding preservation classification remained stable for all 13. This is evidence about the dated, recorded cloud workflow; it does not pin the backend or guarantee future service behavior.

F. Integrated-document sanity check

The secondary sanity check evaluated three project-generated two-page multi-feature documents: I01, a community access report; I02, an educational handout; and I03, a policy note. They used only existing contract families, and a separately implemented raw-OOXML oracle passed 21/21 integrated source assertions. The same two pipelines produced six tagged PDFs, and the protocol generated 42 property–pipeline observations without requiring a document-level score. The secondary result distribution was 16 VERIFIED_PRESERVED, 6 OBSERVED_PARTIAL, 5 ALTERED, 3 CONFIRMED_LOST, 11 UNRESOLVED_EQUIVALENCE, and 1 MEASUREMENT_ERROR. These observations demonstrate protocol applicability when properties coexist; they are not a prevalence estimate and are excluded from the primary 52-case atomic denominator.

V. Results

A. Overall accounting

All 52 atomic fixture–pipeline cases produced genuine, parseable, non-empty, structurally tagged PDFs that passed the predefined structural output checks. The atomic evidence-aware counts are 18 VERIFIED PRESERVED, 4 OBSERVED PARTIAL, 7 ALTERED, 4 CONFIRMED LOST, 18 UNRESOLVED EQUIVALENCE, and 1 MEASUREMENT ERROR. These are controlled-case counts, not prevalence estimates or failure rates. The 18 unresolved cases are the frozen primary analysis.

TABLE I
FROZEN FIXTURE CONTRACTS.

Fixture	Property	Contract focus
F01	Headings	Heading levels and order
F02	Alternative text	Author-written image description
F03	Lists	Ordered/unordered type, nesting, and list structure
F04	Table	Table, row/cell, and header structure
F05	Document language	Primary document language value
F06	Inline language	Run-level language override within an en-US document
F07	Decorative image	Meaningful versus explicitly decorative image state
F08	Hyperlinks	Link existence, visible text, and URI destination
F09	Document title	Visible title and document metadata identity
F10	Complex table	Multi-level headers, spans, and associations
F11	Footnotes	Note body and reference-to-note association
F12	Equation	Native equation representation and visible expression
F14	Caption	Figure/caption text and association

A separate additive hyperlink-evidence audit deterministically resolved four of those rows, leaving 14 unresolved in the derived analysis; it did not overwrite the frozen result file or primary denominators. The earlier Core denominator remains 26 cases and is reported separately in the provenance records.

B. Feature-family matrix

Table II is the central atomic matrix. "Preserved" denotes VERIFIED PRESERVED; "Partial" denotes OBSERVED PARTIAL; "Lost" denotes CONFIRMED LOST; "Unresolved" denotes UNRESOLVED EQUIVALENCE; and "Meas." denotes MEASUREMENT ERROR. The Core and Variant columns are separate controlled instances, not repeated samples from a population.

The feature matrix shows why a single converter score would be misleading. The same named pipeline can preserve one feature and alter another, and unresolved equivalence is concentrated in mappings where the destination representation is present but not fully comparable. The counts are therefore secondary to the feature-level evidence and should not be read as a global accessibility rate.

C. Confirmed losses and difficult mappings

The four confirmed-loss rows are the Core and Variant B instances of F06/B06 inline language and F11/B11 footnote association under the Google Docs DOCX-to-PDF conversion pipeline. In the inline-language cases, the source contracts require non-default run-level language values. The pypdf structure traversal and PyMuPDF serialized-object scan found only document-level language and no required inline-language span or language markers; visible text remained. In the footnote cases, visible reference markers and note text remained, but both paths found no machine-verifiable `/Note`, `/Footnote`, `/Reference`, or equivalent association. These are losses of the contracted machine-verifiable associations, not claims about complete PDF accessibility or user harm.

The two-path audit also explains the unresolved rows. For list, hyperlink, complex-table, equation, and caption mappings, destination structure or visible content often survives, but the available evidence does not establish a feature-specific equivalent representation. The comparator therefore abstains. Absence from one extracted encoding is not treated as semantic absence when a valid alternate representation remains possible.

As a deliberately weaker contrast, a destination-presence-only baseline found a feature-shaped destination structure in 16 of the 18 frozen unresolved rows. This does not resolve those cases: destination presence alone does not establish preservation of the corresponding source semantic relationship.

D. Google repeatability

The separate Core repeatability arm contains 13 fixtures and 39 total Google runs. All 13 fixtures were classification-stable across Run 1, Run 2, and Run 3; all three PDFs were byte-identical for every fixture under the recorded workflow. This is a dated cloud-workflow observation. The backend version cannot be pinned, and the result does not guarantee future Google Docs behavior.

E. Integrated-document sanity check

The separate secondary evaluation used three project-generated multi-feature documents and 42 property–pipeline observations, after 21/21 source-oracle assertions passed. Those observations are retained as a protocol-applicability sanity check; they are not merged with the 52 atomic cases, do not create a representative corpus, and do not estimate real-world prevalence.

VI. DISCUSSION

A. What the benchmark demonstrates

The main result is methodological: accessibility-information preservation can be tested when source properties are explicitly known, destination representations can be inspected, and the oracle is allowed to abstain. The 52-case atomic benchmark contains 18 verified preserved cases, 4 observed partial cases, 7 alterations, 4 confirmed losses, 18 unresolved equivalence cases, and 1 measurement error. The categories do not collapse uncertainty into converter degradation. Directly demonstrated

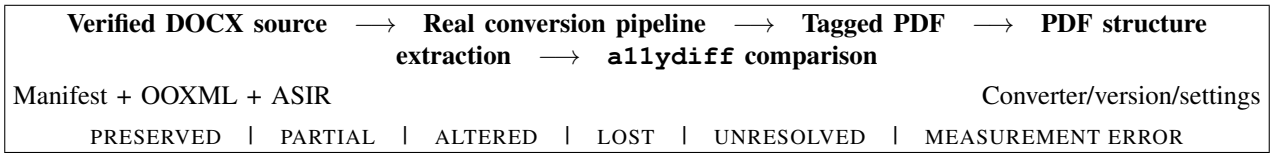


Fig. 1. A11yPreserve's source-grounded measurement workflow. The destination is inspected, but the outcome is defined by comparison with verified source information.

TABLE II
ATOMIC BENCHMARK: TWO CONTROLLED INSTANCES PER SELECTED FEATURE FAMILY, EACH RUN THROUGH BOTH NAMED PIPELINES.

Feature	LO Core	LO Var.	G. Core	G. Var.	Interpretation
Headings	Preserved	Preserved	Partial	Partial	Google adds a generated H1 in both controlled cases
Alt text	Altered	Altered	Preserved	Preserved	Exporter-added wording occurs only in the tested LO cases
Lists	Preserved	Unresolved	Unresolved	Unresolved	List structure survives, but some type/text mappings remain unresolved
Table	Preserved	Unresolved	Preserved	Unresolved	Variant cell correspondence is not established by extraction
Document language	Preserved	Preserved	Altered	Altered	Google reduces locale specificity in both cases
Inline language	Preserved	Preserved	Lost	Lost	Required inline language representation absent in Google rows
Decorative image	Partial	Altered	Partial	Preserved	Decorative-state evidence is fixture-sensitive
Links	Unresolved	Unresolved	Unresolved	Unresolved	URI survives; visible-name association remains unresolved
Document title	Preserved	Preserved	Altered	Altered	Google metadata identity is reduced in both cases
Complex table	Preserved	Unresolved	Unresolved	Unresolved	Multi-level association equivalence remains unresolved
Footnotes	Meas.	Unresolved	Lost	Lost	Google association loss is corroborated; LO evidence varies by fixture
Equation	Preserved	Unresolved	Unresolved	Unresolved	Native expression equivalence is not always measurable
Captions	Preserved	Preserved	Unresolved	Unresolved	Google retains text but not a verified Caption association

changes are separated from cases where an alternate cross-format representation cannot yet be established.

This distinction matters because destination accessibility and source-information preservation answer different questions. A tagged PDF can contain headings, figures, or tables and still fail to preserve a particular source property. Conversely, relevant visible content or a destination structure can remain while the available evidence is insufficient to establish feature-specific equivalence. A11yPreserve operationalizes this difference as a source-contract test oracle rather than as a destination-only accessibility score.

B. Testing contribution and transfer

For a software-testing audience, the pipeline is explicit: contracts are test oracles, controlled DOCX files are test inputs, each conversion pipeline is a system under test, PDF extraction produces observations, and the precedence rules make evidence-aware oracle decisions. Calibration controls expose alternate encodings that the comparator cannot safely resolve; those cases become actionable unresolved results rather than silent false failures or false passes. The transferable lesson is methodological—source-grounded differential testing needs representability-aware contracts, provenance, and explicit oracle uncertainty—not a claim of a general testing theory.

The two controlled instances per feature family show where the protocol is repeatable and where mappings remain difficult. The separate Google repeatability arm strengthens the operational claim for the recorded workflow: all 13 classifications were stable across three runs and all corresponding PDFs were byte-identical. This does not establish backend invariance beyond the dated environment.

C. Pipeline observations

The same source family can produce different outcomes across the two tested pipelines and across feature families. The evidence supports only observations about LibreOffice 26.2.6.3 and the recorded Google Docs DOCX-to-PDF conversion pipeline. It does not establish that Google Docs is generally worse, that LibreOffice is generally better, or that either pipeline represents an ecosystem-wide pattern.

The Google condition includes DOCX import, Google's internal document representation, and PDF generation. The experiment therefore cannot isolate whether a difference arose during import, internal representation, or PDF generation. The Variant B Google PDFs were obtained through the authenticated in-session export backend after the visible download action did not register; this route is disclosed in the conversion report rather than treated as a separately isolated exporter test.

D. Related-work boundary

Existing work already addresses accessible conversion, PDF accessibility benchmarking, destination evaluation, PDF-to-HTML reconstruction, remediation, and accessible document

generation. DocAccessible and DAISY provide particularly relevant conversion-benchmark context [8], [9]; SciA11y, iTagPDF, and prior office-document conversion studies address adjacent transformation or destination problems [4]–[6]. The contribution claimed here is narrower: a controlled, source-grounded, contract-based, uncertainty-aware conformance-testing protocol for machine-verifiable accessibility preservation across the tested DOCX-to-PDF pipelines. It is not a claim to be the first accessible-conversion study, a unique validator, or a replacement for destination-only evaluation.

E. User and engineering implications

The protocol can help authors and publishing engineers regression-test whether selected author-supplied properties survive a known transformation. It does not determine task completion, screen-reader usability, perceived accessibility, or severity for disabled users. An altered value may remain usable, a partial structure may provide substantial accessibility, and an unresolved case may ultimately prove equivalent under a richer representation. Structural evidence should therefore guide review and regression testing, not be presented as a complete accessibility judgment.

VII. Threats to Validity

A. Controlled corpus and adequacy

The 26 atomic fixtures are controlled cases: two project-generated instances for each of 13 selected feature families. They are not a representative sample, and feature families differ in complexity and accessibility relevance. The selection basis is purposive—machine representability, accessibility relevance, conversion relevance, prior structural evidence, and feasible source-to-destination mapping—not a claim of standards completeness. No formal coverage metric, mutation score, or exhaustive fault model is claimed. The integrated documents are also project-generated multi-feature documents and provide applicability evidence rather than external validity.

F13 was reserved for reading order but deferred because reliable automated source-to-PDF equivalence could not be established. This is a concrete example of measurability selection bias: the suite emphasizes properties whose source and destination representations can be inspected under explicit contracts.

B. Converter and format coverage

The atomic benchmark evaluates DOCX-to-PDF only, using LibreOffice and the Google Docs DOCX-to-PDF conversion pipeline. It does not characterize Microsoft Word, ODT, HTML, EPUB, other converters, or the converter ecosystem. The two named pipelines support condition-specific observations, not a global product ranking. Microsoft Word is excluded from the denominator because it was not a completed primary workflow.

C. Cloud reproducibility and route provenance

Google Docs is a cloud service whose backend version cannot necessarily be pinned. The study records the date, operating system, browser, observed renderer, workflow, source hashes, output hashes, and validation evidence. The 39-run repeatability result is therefore conditional on the recorded environment and does not guarantee future service behavior. For Variant B, the visible download action did not register after retries and the authenticated in-session Google export backend supplied the saved PDFs; this deviation is documented and prevents a stronger claim that every atomic arm used an identical UI-download path.

D. Cross-format representation and parser dependence

Source and destination formats express accessibility information differently. The study uses feature-specific contracts and machine-inspectable mappings, such as Word Heading 2 to PDF `/H2`, a Word run-language override to PDF `/Lang`, or a native equation to PDF `/Formula`. Some mappings are incomplete or parser-sensitive. The calibration audit records two false negatives and seven unresolved or indeterminate controls; these limitations motivate the explicit UNRESOLVED EQUIVALENCE category. Absence from one extracted encoding is not automatically proof of semantic absence when a valid alternate representation remains possible.

The primary destination extractor is supplemented by raw object inspection, PyMuPDF checks, and Poppler text extraction for difficult cases. These checks reduce but do not eliminate parser error. `a11ydiff` is not a complete accessibility validator, and conventional validators are not called defective merely because they answer a different destination-only question.

E. Oracle independence

Source ground truth was established from the source manifest, raw OOXML inspection, and source-extractor agreement. A separately implemented raw-OOXML oracle passed all 13 Core contracts and all 13 Variant B contracts, but it was written by the same research team; this is not external expert validation. The ASIR is implementation-grounded and shares the broader toolchain. The independent path is useful cross-checking evidence, not proof of absolute oracle independence.

F. User impact

No disabled-user study, screen-reader task study, or perceived-usability study was performed. The results measure preservation of machine-verifiable accessibility information in destination structure. They do not measure task completion, screen-reader usability, perceived accessibility, or severity for disabled users. Altered, partial, or lost contract information is not automatically equated with user harm.

G. Interpretation and generalization

Counts are descriptive controlled-case summaries. No inferential statistics, prevalence estimate, accessibility rate, failure rate, or weighted converter score is justified. Results do not

establish that document conversion generally destroys accessibility, that Google Docs is generally worse or LibreOffice generally better, that findings generalize to all DOCX documents or software versions, that altered information necessarily causes harm, or that PDF export alone caused every observed change.

VIII. Conclusion

A11yPreserve demonstrates a controlled, source-grounded way to test whether machine-verifiable accessibility information survives document conversion. The atomic benchmark used 26 controlled DOCX fixtures—two instances for each of 13 selected feature families—through LibreOffice and the Google Docs DOCX-to-PDF conversion pipeline, with explicit contracts, source-oracle checks, destination evidence, provenance, and conservative uncertainty handling.

Across 52 atomic cases, the evidence-aware outcomes were 18 verified preserved, 4 observed partial, 7 altered, 4 confirmed lost, 18 unresolved equivalence, and 1 measurement error. A separate 39-run Google Core repeatability study found stable classifications for all 13 fixtures and byte-identical PDFs across the three recorded runs. The integrated sanity check remains separate at three documents and 42 property–pipeline observations, after 21/21 source assertions passed.

The conclusions are intentionally bounded. The fixtures are controlled and purposive rather than representative, the selected features do not provide exhaustive accessibility coverage, only two named DOCX-to-PDF pipelines were evaluated, the Google backend is cloud-versioned, Variant B used a documented in-session export recovery route, and no disabled-user study was performed. The results support a reusable testing protocol and benchmark for selected preservation contracts. They do not support claims about general accessibility, a global product ranking, all DOCX documents, all software versions, universal user harm, or PDF-export-only causation. Future work can extend the corpus with justified variants, compare additional converters, improve cross-format equivalence mappings, and connect structural changes to user tasks without changing the central question: did known accessibility information survive conversion?

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